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CAP 463: CAPSTONE PROJECT

**TITLE:- HealNet – Web-Based Remote Telemedicine
Platform**

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1. Introduction

Healthcare accessibility remains a significant challenge in many regions due to geographical barriers, limited specialist availability, and time constraints. With the rapid growth of web technologies, healthcare services can now be delivered efficiently through secure online platforms.

HealNet is a web-based remote telemedicine platform designed to integrate online medical consultation, intelligent disease search, and hospital location services into a single unified system. The platform aims to improve healthcare accessibility by allowing users to consult doctors, search diseases, and locate hospitals from anywhere.

HealNet supports digital healthcare initiatives by providing a technology-driven solution for efficient and convenient medical services.

2. Problem Statement

Patients often experience difficulties in:

- Identifying the correct specialist for a particular disease
- Finding nearby hospitals quickly
- Booking timely medical consultations
- Managing medical records securely
- Accessing healthcare information remotely

Traditional healthcare systems depend heavily on physical visits and manual processes. A centralized web-based platform like **HealNet** can improve accessibility, efficiency, and patient convenience.

3. Aim of the Project

To design and develop **HealNet**, a secure, scalable, and user-friendly web-based telemedicine platform that enables disease search, hospital discovery, and online consultation through a single digital interface.

4. Objectives

- To develop a fully functional telemedicine web application (HealNet)
- To implement a disease search feature that provides healthcare recommendations
- To integrate a location-based hospital finder using web APIs
- To enable secure online consultation between doctors and patients
- To implement Electronic Medical Record (EMR) management
- To ensure data privacy through authentication and encryption mechanisms

5. Proposed Web System Overview

HealNet will be developed using a **client-server architecture**, where users access services through a web browser.

Intelligent Disease Search and Hospital Locator

When a user searches for a disease or symptom:

- The system displays basic information about the condition
- Relevant medical specialists are identified
- Nearby hospitals are fetched using location-based services
- Available doctors are displayed for consultation

- Results can be filtered by specialization, distance, and availability

This feature helps users make informed healthcare decisions and obtain immediate medical assistance.

6. System Architecture

HealNet uses a **three-tier architecture**:

- Presentation Layer (Angular Frontend)
- Application Layer (Node.js Backend)
- Data Layer (MongoDB Atlas Database)

Overall Architecture Diagram

```
graph TD; Users["Users (Patient / Doctor / Admin)"] --> Angular["Angular Frontend"]; Angular --> REST["REST APIs"]; REST --> Node["Node.js + Express"]; Node --> Database["Database + External APIs"];
```

Doctor Module Architecture

Doctor
↓
Angular Interface
↓
Login
↓
Dashboard
↓

Appointments
Consultation Chat
Prescription
Patient Reports
Availability Status

↓
Node.js Backend
↓
MongoDB Atlas

Patient Module Architecture

Patient



Angular Interface



Login/Register



Dashboard



Disease Search

Symptom Checker

Hospital Locator

Emergency Mode

Appointments

Report Upload



Node.js Backend



MongoDB Atlas

Admin Module Architecture

Admin



Angular Interface



Login



Dashboard



Doctor Verification

User Management

Hospital Management

System Monitoring



Node.js Backend



MongoDB Atlas

Advanced Features Architecture

Symptom Checker System

Users enter symptoms to get disease suggestions.

Symptoms Input



Angular Form



Node.js Processing



Disease Matching



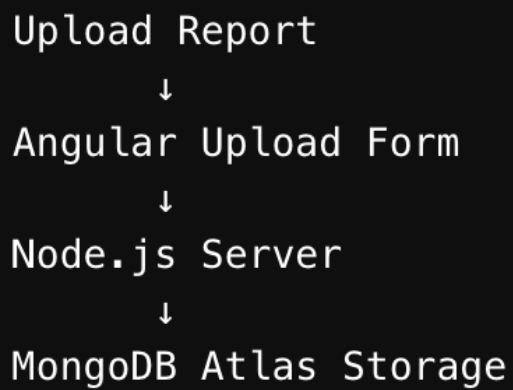
Suggested Specialist

Output:

- Possible diseases
- Recommended doctors

Medical Report Upload

Users upload reports for doctors to review.



7. System Modules

1. Patient Module

- Registration and Login
- Disease Search
- Symptom Checker
- Hospital Locator
- Appointment Booking
- Emergency Mode

- Medical Report Upload
- Medical History Management

2. Doctor Module

- Doctor Login
- Appointment Management
- Online Consultation
- Prescription Generation
- Patient Record Access
- Availability Status

3. Admin Module

- User Management
- Doctor Verification
- Hospital Management
- System Monitoring

4. Hospital Locator Module

- Location Detection via Browser

- Google Maps Integration
- Distance-Based Hospital Listing
- Hospital Specialization Display

8. Technology Stack

Frontend

- Angular
- TypeScript
- HTML5
- CSS3

Backend

- Node.js
- Express.js

Database

- **MongoDB Atlas (Cloud Database)**

MongoDB Atlas is used to store:

- Patient Data

- Doctor Data
- Appointments
- Medical Records
- Hospital Information

APIs Used

1. Google Maps API

Used for:

- Hospital location services
- Distance calculation
- Map display

2. Browser Geolocation API

Used for:

- Detecting user location
- Providing coordinates for hospital search

3. Socket.IO

Used for:

- Real-time consultation chat between doctors and patients

4. Multer API

Used for:

- Uploading medical reports and prescriptions

5. JWT Authentication

Used for:

- Secure login
- User authentication
- Role-based access control

9. Security Features

- HTTPS Protocol
- Password Encryption (bcrypt)
- JWT Authentication
- Secure APIs
- Role-Based Access Control

10. Expected Outcome

HealNet will result in a responsive and secure web application capable of:

- Providing online medical consultation
- Recommending nearby hospitals
- Managing digital medical records
- Improving healthcare accessibility

The integration of disease search and hospital mapping improves usability and innovation.

11. Future Scope

- AI-based symptom prediction
- Integration with wearable health devices
- Emergency ambulance tracking
- AI chatbot for preliminary medical guidance
- Multi-language support

12. Conclusion

HealNet is a comprehensive web-based telemedicine platform that integrates teleconsultation, hospital discovery, and secure medical